SysMon: System Monitor for Private Infrastructure

Requirements and System Design Specification

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1 Project Overview

Project Title: SysMon
Target Audience: Homelab enthusiasts and small-sized businesses wanting to utilize their own infrastructure.
Goal: To deliver a seamless user experience similar to Web UIs from enterprise solutions (Azure/AWS), allowing both technical and non-technical users to monitor, maintain, and build on private infrastructure.

1.1 Core Modules

1. **Frontend:** User Interface for visualization, configuration, and monitoring.
2. **Backend:** Orchestration of tasks, container management, telemetry aggregation, and host-level filesystem management.
3. **Authentication System:** Tier-based access control (Tier 1 to Tier 3) for secure team collaboration.

2 Key Features

- **Container Orchestration:** Support for both Docker and Podman. Features manual container creation via Compose files, alongside a visual UI builder that auto-generates Compose files.
- **Resource Governance (Resource Groups):** Tier 1 admins can create profiles limiting compute resources (e.g., thread limits, RAM caps, GPU access restrictions) enforced at the cgroup level via Compose deployments.
- **Storage Quotas (Storage Groups):** Provisioning of bounded host directories for container bind-mounts. Ensures specific user groups cannot exceed predetermined disk space limits.
- **Network Management:** Creating profiles, static/dynamic IP allocation, and a firewall rules manager.
- **Telemetry & Monitoring:** Real-time visualization of Memory utilization, CPU utilization, and Disk space.

2.1 Feature priorities

- **Must have:**
  - **Support for either Docker or Podman:** SysMon is meant to act as a wrapper of a different system, equating support with basic functionality.
  - **Functional Resource Groups:** Enforceable resource access rules are mandatory in a team-based environment.
  - **Functional Storage Groups:** As with Resource Groups, lacking this functionality introduces potential security risks.

- **Should have:**
  - **Support for both Docker and Podman:** Versatility is supposed to be SysMon's main selling point - supporting both tools delivers on it.
  - **Container creation through Compose files:** Requiring the user to access the underlying system directly to create containers is counterintuitive at best and should be avoided.
  - **Network Management Capabilities:** No container networking support would heavily restrict the variety of tasks SysMon can perform and, given how prevalent such methods are, shrink the already limited target audience.
- **Could have:**
  - **Telemetry Data Visualization:** While not mandatory per se, the inclusion of an appropriate monitoring panel would greatly enhance user comfort.

3 User Stories

- **US1 (Resource Governance):** As a Tier 1 admin, I want to create a "Basic Resource Group" that restricts deployments to 2 CPU cores and 4GB of RAM, so that lower-tier users cannot accidentally exhaust the host server's resources.
- **US2 (Storage Quotas):** As a Tier 1 admin, I want to create a Storage Group with a 50GB quota, so that users deploying applications via the visual builder can only map volumes to this restricted folder without filling up the host system's root drive.
- **US3 (Visual Deployment):** As a Tier 3 (newer) admin, I want to visually attach a Docker image and select a pre-approved Resource Group and Storage Group, so that the system automatically generates a compliant Compose file.
- **US4 (Advanced Deployment):** As a Tier 1 admin, I want to paste my existing custom `docker-compose.yml` files directly into the system to easily migrate existing complex homelab stacks.

4 Crucial Use Case Description (Alistair Cockburn's Schema)

Use Case Name	**Visual Configuration and Deployment with Resource Limits**
Primary Actor	Admin (Tier 2 or Tier 3)
Scope	SysMon Orchestration & Frontend Module
Level	User Goal
Brief Description	The admin configures a new container via the UI, assigning it to a specific Resource Group and Storage Group. SysMon generates a Compose file injecting these limits and deploys it.
Preconditions	The admin is authenticated. Resource Groups and Storage Groups have been previously defined by a Tier 1 admin.
Success Guarantee	A Compose file is generated containing `deploy.resources` limits, volume paths are validated against the Storage Group, and the container is running.
Main Success Scenario	1. Admin navigates to "Create Container" and selects "Visual Builder". 2. Admin inputs the image repository tag. 3. Admin selects an assigned Resource Group 4. Admin maps persistent data to an assigned Storage Group folder. 5. Admin selects the target engine (Docker or Podman) and clicks Deploy. 6. SysMon Backend injects cgroup limits and approved host paths into a generated `docker-compose.yml` string. 7. SysMon executes the deployment via the API. 8. System begins streaming telemetry for the new container.
Extensions (Alternative Flows)	6a. Storage Group quota exceeded: System halts generation and notifies the admin that the selected storage pool is full. 7a. Engine daemon unreachable: System saves the generated Compose file as a draft and notifies the admin.

Table 1: Use Case: Visual Configuration with Governance

5 Class Diagram Specification

The following diagram presents the conceptual class structures focusing on visual builder configurations, Compose deployments, and resource/storage governance.

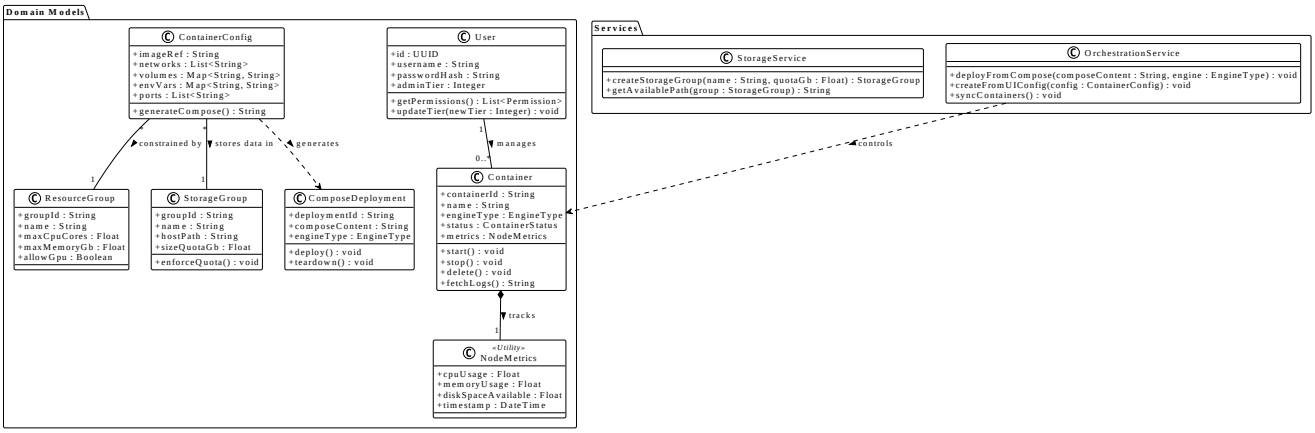


Figure 1: SysMon Domain and Service Class Diagram

Use Case Name	**Creating New Users**
Primary Actor	Admin (Tier 1)
Scope	Authentication System & Frontend Module
Level	User Goal
Brief Description	The admin adds a new user.
Preconditions	The admin is authenticated. There exists a SysMon environment.
Success Condition	The new account is created and accessible.
Failure Condition	No new user account is created.
Main Success Scenario	1. The admin opens the "Users" panel. 2. The admin opens the "Add User" panel. 3. They specify the name, password, and tier of the new account. 4. They commit the introduced changes. 5. The system creates the new user with no permissions.

Table 2: Use Case: Creating New Users

Use Case Name	**Deleting Users**
Primary Actor	Admin (Tier 1)
Scope	Authentication System & Frontend Module
Level	User Goal
Brief Description	The admin deletes a user account.
Preconditions	The admin is authenticated. There exists a SysMon environment and the user account in question. The "Users" panel is open and the "Manage Access" option and the appropriate user are selected.
Success Condition	The user account is deleted.
Failure Condition	The user account persists within the environment.
Main Success Scenario	1. The admin selects the "Delete User" option, opening the confirmation panel. 2. They enter their password and select "Confirm". 3. The user account is deleted by the system.
Extensions	2a. They instead cancel the operation, leading to no changes being done.

Table 3: Use Case: Creating New Users

Use Case Name	**Managing User Access**
Primary Actor	Admin (Tier 1)
Scope	Authentication System & Frontend Module
Level	User Goal
Brief Description	The admin grants or revokes the permission to manage or use a container or group, or removes the user entirely.
Preconditions	The admin is authenticated. There exists a SysMon environment and the appropriate container or group.
Success Condition	The changes become functional the moment they are accepted.
Failure Condition	No changes are introduced.
Main Success Scenario	1. The admin opens the "Users" panel. 2. The admin selects the "Manage Access" option. 3. They select the user, whose permissions they wish to change. 4. They select the desired container. 5. They grant the appropriate permissions to the selected user within the context of the chosen container. 6. They commit the introduced changes. 7. SysMon modifies the properties of the chosen user.
Extensions	4a. They instead select the "Delete User" option (use case 3).
Variations	4. The admin may select one of: container, resource group, storage group. 5. The admin may modify the permissions to: see, use, modify, delete, and create containers and groups.

Table 4: Use Case: Managing User Access

Use Case Name	**Creating Resource and Storage Groups**
Primary Actor	Admin (Tier 1 and 2)
Scope	Backend/Orchestration & Frontend Module
Level	User Goal
Brief Description	The admin creates a resource or storage group with a custom configuration.
Preconditions	The admin is authenticated and has the necessary permissions. There exists a SysMon environment.
Success Condition	The group is successfully created.
Failure Condition	No group is created.
Main Success Scenario	1. The admin opens the "Manage Groups" panel. 2. They select the "Create New Resource Group" option. 3. They specify the RAM, CPU, and GPU restrictions enforced within the group. 4. The system successfully verifies the validity of the given configuration, ensuring the given values do not exceed hardware capabilities. 5. They commit the introduced changes. 6. SysMon creates the new group according to the provided specification.
Extensions	2a. They select the "Create New Storage Group" option. 3a. They specify size quota enforced within the group (step 2a is a prerequisite). 4a. The system rejects the provided configuration (leads back to step 3/3a).

Table 5: Use Case: Creating Resource and Storage Groups

Use Case Name	**Deleting Resource and Storage Groups**
Primary Actor	Admin (Tier 1 and 2)
Scope	Backend/Orchestration & Frontend Module
Level	User Goal
Brief Description	The admin deletes an unused resource or storage group.
Preconditions	The admin is authenticated and has the necessary permissions. There exists a SysMon environment and the appropriate group.
Success Condition	The group is successfully deleted.
Failure Condition	No group is deleted.
Main Success Scenario	1. The admin opens the "Manage Groups" panel. 2. They select the group they wish to delete. 3. They select the "Delete Group" option, causing a confirmation request. 4. The system successfully verifies that the chosen group remains unused. 5. The admin selects the "Confirm" option. 6. SysMon deletes the chosen group.
Extensions	4a. The system rejects the provided configuration (leads back to step 2).
Variations	2. The admin may select either of: resource group, storage group.

Table 6: Use Case: Deleting Resource and Storage Groups

Use Case Name	**Managing Existing Resource and Storage Groups**
Primary Actor	Admin (Tier 1 and 2)
Scope	Backend/Orchestration & Frontend Module
Level	User Goal
Brief Description	The admin modifies the properties of a chosen resource or storage group.
Preconditions	The admin is authenticated and has the necessary permissions. There exists a SysMon environment and the appropriate resource or storage group.
Success Condition	The changes become functional the moment they are accepted.
Failure Condition	No changes are introduced.
Main Success Scenario	1. The admin opens the "Manage Groups" panel. 2. They select the group they wish to modify. 3. They introduce and confirm the desired changes. 4. The system successfully verifies the new configuration, ensuring the given values do not exceed hardware capabilities. 5. The admin commits the introduced changes. 6. SysMon introduces the change into the system.
Extensions	5a. The system rejects the provided configuration (leads back to step 3).
Variations	2. The admin may select one of: resource group, storage group. 3. The changes may concern RAM, CPU, and GPU restrictions for resource groups and quotas for storage groups.

Table 7: Use Case: Managing Existing Resource and Storage Groups